

Archaeological Explanation

Archaeology is generally considered to be a social science: a discipline of using systematic approaches to study people and their societies. Archaeology is distinct from other social sciences in that archaeologists study the past, which makes it impossible to directly question people about their behavior or observe their actions. Instead, archaeologists use modes of **inference** to make educated guesses about the way that certain activities or social processes worked in the past, and then find **evidence** to test those inferences.

Archaeological Inference

Inference is the key mechanism by which archaeologists attempt to explain the past. The process of archaeological inference uses prior information (whether from prior archaeological research or from another discipline or line of evidence) to put forward an explanation of a phenomenon or relationship that is possible to identify using archaeological methods and forms of evidence. The evidence is then collected and analyzed, and results are compared with the original inference, either lending support to that explanation or suggesting that another explanation is more likely. This approach shares similarities with the so-called “scientific approach” of hypothesis testing. The difference is that archaeologists are not able to directly test hypotheses because our evidence is only ever a **proxy**, or a substitute, for the real data we seek.

For example, if we hypothesized that Americans of Thai descent ate less meat than Americans of Chinese descent because a greater percentage of Thai (and Thai Americans) were Buddhist, we could test this proposition directly by surveying, interviewing, or observing both populations and comparing the results. But if archaeologists were investigating an ancient city in Southeast Asia and attempting to determine whether the people who lived there were Buddhist, they might make the inference from finding many animal bones that the population was not Buddhist, or from finding very few animal bones, that they were Buddhist. But the presence of animal bones is only a **proxy** of the real piece of evidence the archaeologists hope to recover: the religion of the inhabitants.

Inference is the use of observation and evidence to establish a conclusion. Archaeological inference draws on

archaeological evidence to make statements about likely conditions in the past.

A **proxy** is a substitute. In the case of archaeological and paleoenvironmental reasoning, proxies are types of evidence that are argued to be similar to, or a direct consequence of, another form of evidence, or a behavior, that we cannot directly observe.

Archaeological Evidence

Archaeological evidence comes in many forms. Archaeologists often use material remains as evidence to test their inferences. They may also use representations of the past from art or written texts. In some cases, archaeologists may rely just as much on these types of evidence as on material remains. An example is the field known as **historical archaeology**, which refers to the archaeology of the Americas since the arrival of Europeans. Since we have written records (in English, Spanish, French, Dutch, or Portuguese, depending on the country and time period) that describe many details of life in these periods, historical archaeologists may spend as much time combing through archives for letters, probate records, land surveys, and tax records as they do analyzing artifacts or exploring archaeological sites. Archaeologists must also be trained in historical methods, and their published reports may rely just as much on the study of written records as the study of material remains.

Archaeological evidence includes material remains, **features**, the spatial relationships among them, imagery and texts from archaeological settings, and results from the scientific analysis of material remains.

Features are a category of immobile material remains. Tools, pots, and jewelry are artifacts, not features, because they are portable. Remains of house walls, carved stone facades built into a natural rock

outcrop, patches of burned earth where fireplaces once sat, and even pits are examples of features.

Historical archaeology generally refers to the field of study of the colonial period (post-1492) in the Americas. In this period, historical documents, as well as archaeological data, are available as evidence.